

ALEXANDRIA, MN, USA

WMO: 726557

Lat: 45.868N

Lon: 95.394W

Elev: 432

StdP: 96.25

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14910

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.9	-31.0	0.2	-27.1	-28.3	0.3	-24.3	12.6	-10.2	11.5	-8.5	3.6	290	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.1	22.3	29.4	21.3	27.9	20.5	24.1	28.9	22.9	27.5	21.8	26.2	5.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.5	18.2	27.3	21.4	16.9	26.0	20.2	15.7	24.7	75.0	28.9	70.0	27.4	65.7	26.3	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.6	10.4	9.1	DB	-30.8	33.9	3.3	1.7	-33.1	35.1	-35.0	36.1	-36.9	37.0	-39.3	38.2	(4)
(5)			WB	-30.7	25.9	2.8	1.1	-32.7	26.6	-34.4	27.3	-35.9	27.9	-38.0	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.1	-11.8	-9.6	-2.3	6.1	13.3	19.0	21.7	20.3	15.9	7.9	-0.6	-8.2	(6)	
	DBStd	13.01	7.82	7.24	7.25	5.76	4.83	3.77	3.12	2.97	4.63	5.10	6.12	6.84	(7)	
	HDD10.0	2764	676	549	385	142	22	1	0	0	6	101	321	563	(8)	
	HDD18.3	4764	934	782	641	368	168	36	8	14	101	324	568	822	(9)	
	CDD10.0	1326	0	0	3	24	125	269	362	321	182	37	3	0	(10)	
	CDD18.3	285	0	0	0	1	13	55	112	76	27	2	0	0	(11)	
	CDH23.3	2296	0	0	0	11	151	447	911	544	213	19	0	0	(12)	
	CDH26.7	614	0	0	0	2	44	119	267	130	49	3	0	0	(13)	
(14) Wind	WSAvg	4.7	4.7	4.7	4.8	5.2	5.0	4.3	4.0	3.9	4.5	5.0	4.9	4.7	(14)	
(15) Precipitation	PrecAvg	649	22	15	34	59	75	110	79	83	63	54	30	17	(15)	
	PrecMax	1017	106	55	77	165	150	277	202	202	138	208	103	66	(16)	
	PrecMin	290	0	1	5	0	7	14	20	5	11	0	1	2	(17)	
	PrecStd	156	21	11	19	35	34	58	42	53	37	50	27	14	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	8.6	18.2	24.5	31.4	33.1	33.5	32.2	30.6	26.2	17.6	7.7	(19)	
		MCWB	2.4	6.3	12.3	15.0	18.7	22.5	24.4	23.0	21.4	16.9	11.9	4.7	(20)	
	2%	DB	2.4	4.7	13.9	20.7	27.4	29.7	31.2	29.8	27.9	21.6	13.6	4.0	(21)	
		MCWB	0.5	2.8	9.3	11.6	17.6	21.1	23.2	21.9	20.2	14.7	8.7	1.8	(22)	
	5%	DB	0.6	2.3	10.1	18.1	24.4	27.6	29.5	28.0	25.9	18.2	10.3	2.0	(23)	
		MCWB	-0.4	1.0	6.8	10.3	15.7	19.7	21.9	20.8	18.9	12.9	7.0	0.5	(24)	
	10%	DB	-1.1	0.7	6.8	15.4	21.8	25.9	27.9	26.5	23.6	15.7	7.7	0.4	(25)	
		MCWB	-1.9	-0.2	4.5	9.1	14.5	18.7	21.1	19.9	17.9	11.4	5.0	-0.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.7	6.2	13.8	15.7	21.1	24.4	26.1	25.1	23.3	18.0	13.2	5.3	(27)	
		MCDB	5.3	8.5	16.5	23.0	27.9	30.0	30.9	30.3	27.9	22.5	16.4	7.1	(28)	
	2%	WB	0.8	2.9	10.0	13.1	18.8	22.6	24.7	23.3	21.6	15.9	9.6	2.1	(29)	
		MCDB	1.9	4.5	13.0	18.1	24.8	27.8	29.6	27.6	26.1	20.1	12.8	3.6	(30)	
	5%	WB	-0.4	1.1	6.9	11.1	17.0	21.2	23.4	22.2	20.3	13.7	7.2	0.7	(31)	
		MCDB	0.6	2.1	9.9	16.9	22.5	26.3	28.1	26.2	24.3	17.6	10.1	1.8	(32)	
	10%	WB	-2.0	-0.4	4.4	9.5	15.3	19.8	22.2	21.1	18.9	11.8	5.1	-0.5	(33)	
		MCDB	-1.1	0.7	6.9	14.7	20.5	24.4	26.4	25.0	22.5	14.9	7.8	0.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.8	9.2	9.0	10.3	10.8	10.1	10.1	10.2	10.3	9.0	7.7	7.6	(35)	
		MCDBR	9.3	9.9	12.6	15.1	15.1	12.3	11.5	11.9	12.9	13.2	11.8	9.5	(36)	
	5% WB	MCWBR	7.9	8.3	8.4	8.3	7.5	6.0	5.6	5.8	6.6	7.5	7.8	7.7	(37)	
		MCWBR	8.7	9.3	12.2	13.7	12.9	10.8	10.7	10.5	10.8	11.7	11.1	8.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.262	0.285	0.305	0.356	0.381	0.382	0.380	0.387	0.354	0.308	0.278	0.255	(40)	
		taud	2.397	2.335	2.404	2.337	2.326	2.372	2.377	2.355	2.432	2.543	2.528	2.443	(41)	
	Ebn at Noon		857	897	928	902	886	885	881	860	863	866	838	830	(42)	
		Edh at Noon	70	92	100	118	123	118	116	115	97	75	62	61	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.68	2.78	3.81	4.76	5.39	5.96	6.37	5.32	4.04	2.53	1.60	1.17	(44)	
	RadStd		0.21	0.31	0.36	0.40	0.41	0.39	0.24	0.45	0.30	0.37	0.16	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page